

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To design a BIST architecture using a Linear Feedback Shift Register (LFSR).	
Experiment No: 12	Date: 10-04-2026	Enrolment No: 92301733056

AIM: To design a BIST architecture using a Linear Feedback Shift Register (LFSR).

IDE: EDA Playground

THEORY:

A Linear Feedback Shift Register (LFSR) is a type of shift register where the input bit is determined by a linear function of the previous states of the register. In essence, it's a chain of storage elements (like flip-flops) connected in a specific way that produces a sequence of output bits. This sequence can be used for various purposes, including pseudo-random number generation, data scrambling, and cryptography.

Here's a more detailed explanation:

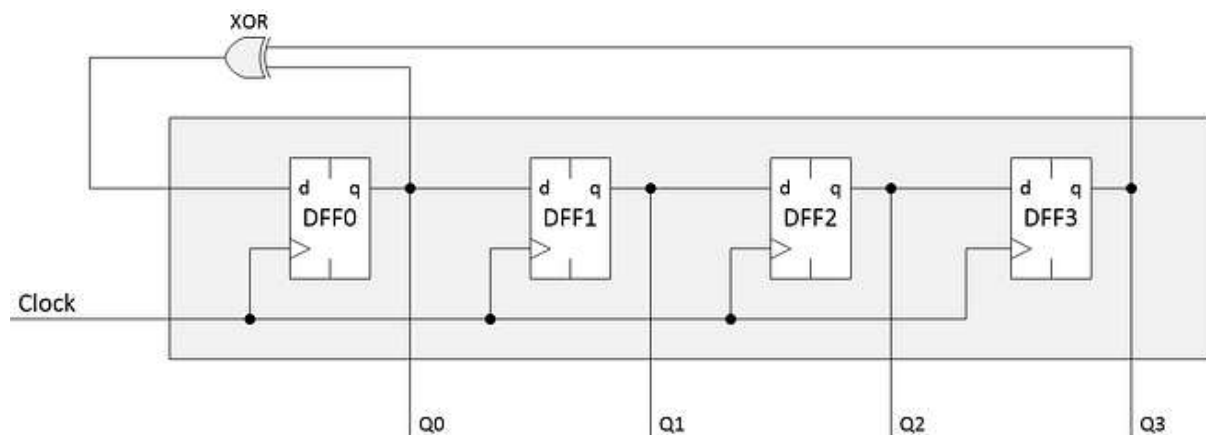
- **Shift Register:**

An LFSR is essentially a shift register, meaning it stores bits and shifts them from one stage to the next on each clock cycle.

- **Linear Feedback:**

The key feature is the feedback mechanism. The input bit for the next stage is not simply the output of the last stage; instead, it's a linear function of some of the previous states (often XORed together).

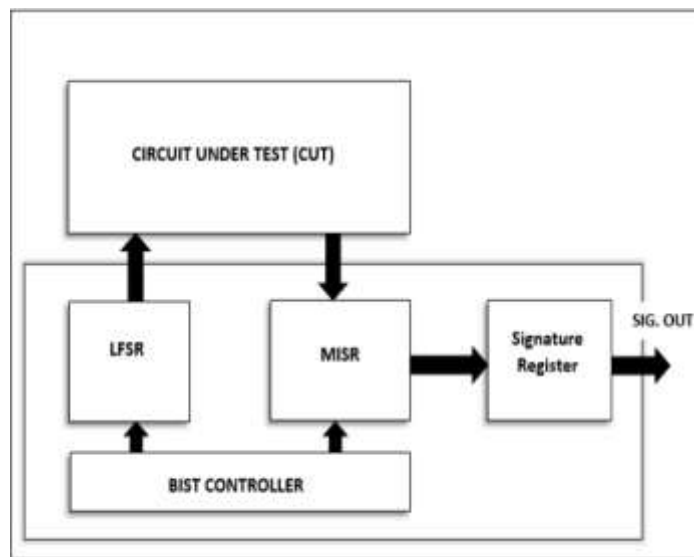
Linear feedback shift registers make extremely good pseudorandom pattern generators. When the outputs of the flip-flops are loaded with a seed value (anything except all 0s, which would cause the LFSR to produce all 0 patterns) and when the LFSR is clocked, it will generate a pseudorandom pattern of 1s and 0s. Note that the only signal necessary to generate the test patterns is the clock.



In simpler terms: Imagine a row of light bulbs. Each bulb can be either on (1) or off (0). The LFSR shifts the state of the bulbs down the row on each clock cycle. But before the shift, the input to the first bulb is determined by a rule based on the current states of some of the previous bulbs (e.g., the first bulb's input is the XOR of the second and fourth bulb's states). This rule creates a sequence of outputs.

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Built-In Self-Test (BIST) is a design technique used to embed the test logic within a digital circuit, allowing it to test itself. This is especially useful in VLSI systems where external test access is limited. In this experiment, we demonstrate the design of a BIST system using a 4-bit Linear Feedback Shift Register (LFSR) as a test pattern generator, and a simple signature analyzer to validate the outputs of a mock DUT (Design Under Test). The design is simulated using System Verilog and verified on EDA Playground.



As the complexity of VLSI circuits increases, ensuring testability becomes more difficult and costly using traditional external testers. BIST offers an efficient way to verify functionality during design, production, and operation. A typical BIST system includes a test pattern generator (like an LFSR), a circuit under test (CUT), and a response analyzer (signature analyzer).

SYSTEM ARCHITECTURE:

The BIST system implemented in this experiment consists of:

1. LFSR (Linear Feedback Shift Register): Generates pseudorandom test patterns.
2. DUT (Design Under Test): A simple logic circuit used for demonstration (bitwise inverter).
3. Signature Analyzer: Collects DUT outputs and compacts them into a signature using XOR.

DESIGN AND IMPLEMENTATION:

A. LFSR Module

A 4-bit LFSR is built using D flip-flops and XOR feedback from bit 4 and bit 1.

B. DUT Module

A simple 4-bit inverter that outputs the bitwise complement of the input from LFSR.

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C. BIST Wrapper

This module integrates the LFSR and DUT and computes a final signature using XOR accumulation.

D. Simulation Platform

The entire design is written in SystemVerilog and simulated using the EDA Playground for waveform analysis.

SYSTEMVERILOG CODE:

A. LFSR Module

```

module LFSR_4bit(
    input logic clk,
    input logic rst,
    output logic [3:0] lfsr_out
);

always @(posedge clk or posedge rst)
begin
    if(rst)
        lfsr_out <= 4'b1011;
    else
        lfsr_out <= {lfsr_out[2:0], lfsr_out[3]^lfsr_out[0]};
end

endmodule

```

B. DUT Module

```

module dut(
    input logic [3:0] data_in,
    output logic [3:0] data_out
);
    assign data_out = ~data_in;
endmodule

```

C. BIST Wrapper Module

```

module bist(
    input logic clk,
    input logic rst,
    output logic [3:0] signature
);
    logic [3:0] lfsr_out, dut_out;
    LFSR_4bit lfsr (.clk(clk), .rst(rst), .lfsr_out(lfsr_out));
    dut d (.data_in(lfsr_out), .data_out(dut_out));

    always_ff @(posedge clk or posedge rst) begin if (rst)

```

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```
signature <= 4'b0000;
else
signature <= signature ^ dut_out;
end endmodule
```

D. Testbench

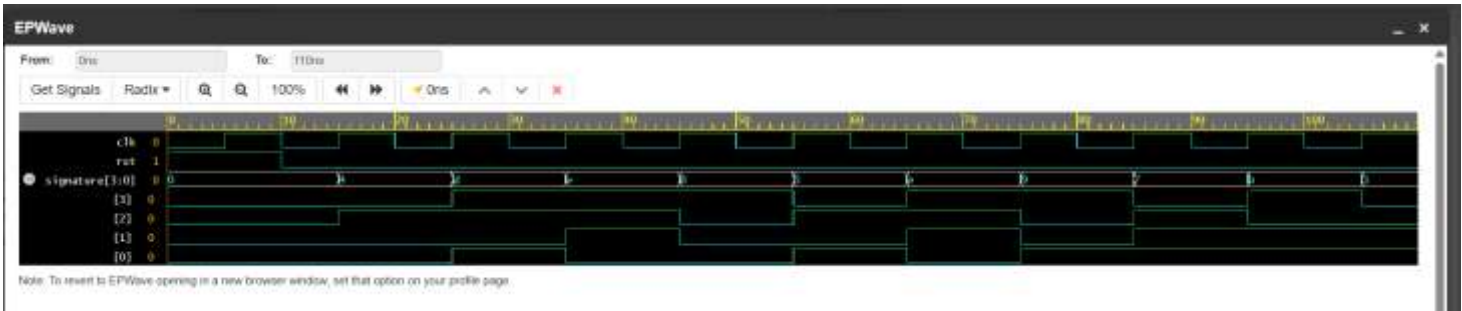
```
// Code your testbench here
// or browse Examples
module test_bist;
logic clk = 0, rst; logic [3:0] signature;

bist uut (.clk(clk), .rst(rst), .signature(signature)); always #5 clk = ~clk;
initial begin
$dumpfile("bist.vcd");
$dumppvars(0, test_bist);

rst = 1; #10;
rst = 0; #100;

$display("Final Signature: %b", signature);
$finish;
end endmodule
```

SIMULATION RESULTS:



CONCLUSION:

In this Experiment I have performed on BIST, the BIST (Built-In Self-Test) architecture using a Linear Feedback Shift Register (LFSR) provides an efficient and reliable method for testing digital circuits. The LFSR generates pseudo-random test patterns that help in detecting faults within the circuit without the need for external testing equipment. This approach reduces testing time, hardware complexity, and overall cost. Additionally, BIST improves fault coverage and enables on-chip testing, making it highly suitable for modern VLSI systems. Thus, the implementation of LFSR-based BIST enhances the performance, reliability, and testability of digital designs.

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Post Lab Exercise:

1. What is Built-In Self-Test (BIST)?

Built-In Self-Test (BIST) is a testing technique in which additional hardware is embedded داخل a digital circuit so that it can test itself. It eliminates the need for expensive external testing equipment by allowing the system to generate test patterns and analyze responses internally.

2. What is the main objective of BIST in digital systems?

The main objective of BIST is to improve fault detection while reducing testing time and cost. It helps in identifying manufacturing defects and operational faults quickly, and ensures high reliability of digital systems without relying heavily on external testers.

3. Define Design for Testability (DFT).

Design for Testability (DFT) is a design approach that simplifies the testing process of a circuit. It involves adding features such as scan chains and test points to improve the controllability and observability of internal signals, making faults easier to detect.

4. What are the key components of a BIST architecture?

A BIST architecture mainly consists of a Test Pattern Generator (TPG), Circuit Under Test (CUT), Output Response Analyzer (ORA) or Signature Analyzer, and a comparator. These components work together to generate inputs, test the circuit, and verify the output responses.

5. What is a Test Pattern Generator (TPG)?

A Test Pattern Generator is a hardware unit, usually implemented using an LFSR, that produces pseudo-random test vectors. These patterns are applied to the circuit to check its functionality and detect possible faults.

6. What is a Signature Analyzer?

A Signature Analyzer is used to compress the output responses of a circuit into a compact form called a signature. This helps in reducing the amount of data that needs to be analyzed and simplifies fault detection.

7. What is the role of a comparator in BIST?

The comparator compares the obtained signature from the circuit with a precomputed correct (golden) signature. If both match, the circuit is considered fault-free; otherwise, a fault is detected.

8. What is meant by on-chip testing?

On-chip testing refers to performing testing operations within the chip itself using built-in test hardware. It avoids the need for external testing devices and allows faster and more efficient testing.

9. What is scan-based BIST?

Scan-based BIST combines scan chain techniques with BIST hardware. It allows test data to be shifted into flip-flops and outputs to be shifted out, improving fault coverage and making sequential circuits easier to test.

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10. What is self-test mode in circuits?

Self-test mode is a special mode of operation where the circuit temporarily stops its normal function and performs internal testing using BIST components. This mode ensures that the system can verify its own correctness.